

SAR image former — detailed design

The as-built design of the spotlight-mode SAR image-formation processor on PolarFire SoC MPFS250T_ES. This is the reference for *how the thing actually works*: dataflow, numeric contracts, memory map, coherency rules, control interface, and the on-board data path.

Companion documents:

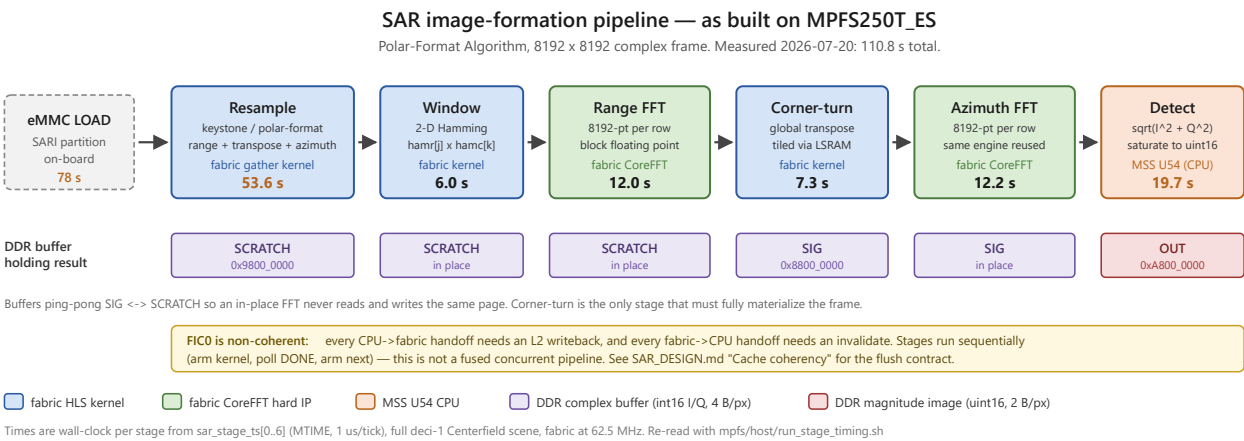
Document	Covers
fpga/SAR_ARCHITECTURE_REPORT.md	Measured per-stage timing and fabric resource usage — the numeric source of truth
fpga/SAR_PIPELINE_STATUS.md	Status, engine history, latency roadmap
fpga/SAR_PIPELINE_PROCESS.md	The PFA math and its host-side golden cross-reference
fpga/SILICON_ISO_TEST_RUNBOOK.md	JTAG bring-up and debug procedure — read before any silicon debug
PROJECT_SOURCE_OF_TRUTH.md	Authoritative index and anti-hallucination rules

1. What it computes

Input is an Umbra CPHD phase-history array (spotlight, X-band, single channel, complex float32). Output is a focused magnitude image. The algorithm is the Polar-Format Algorithm: interpolate the polar-sampled phase history onto a Cartesian k-space grid (the keystone resample), taper it, then take a separable 2-D FFT and detect.

The processing frame is a fixed 8192 x 8192 complex grid. A full capture is decimated to fit; per-axis sizing and rejection rules live in `mpfs/host/ddr_layout.py` (`plan_frame` / `check_input_dims`). Only the native 8192-point transform exists — 16384-point and multi-length paths were dropped.

2. Pipeline



Stages run sequentially: the MSS arms a kernel, polls its DONE flag, then arms the next. This is not a fused concurrent pipeline. Every stage is a DDR-to-DDR streaming pass, because the frame (256 MiB complex) far exceeds on-chip SRAM; on-chip each stage holds only a row, a transpose tile, or AXI burst FIFOs.

#	Stage	Engine	In	Out	Time
1	Resample (range + transpose + azimuth)	fabric gather kernel + MSS coefficients	SIG	SCRATCH	53.6 s
2	Window (2-D Hamming)	fabric kernel	SCRATCH	SCRATCH	6.0 s
3	Range FFT	fabric CoreFFT	SCRATCH	SCRATCH	12.0 s
4	Corner-turn (transpose)	fabric kernel	SCRATCH	SIG	7.3 s
5	Azimuth FFT	fabric CoreFFT	SIG	SIG	12.2 s
6	Detect (magnitude)	MSS U54 CPU	SIG	OUT	19.7 s

Total 110.8 s. Orchestration is `sar_form_image()` in `src/sar/sar_sequencer.c`.

2.0 Kernel-level decomposition

The table above is the *timing* view: six instrumented stages matching `sar_stage_ts[0..6]`. At kernel granularity there are eight arm/wait cycles, because the resample stage internally runs range-resample, a corner-turn, and azimuth-resample before reporting one timestamp. Both views are correct; this is the one to use when reasoning about buffer traffic or arming order.

#	Stage	Kernel	Reads	Writes	Timing stage
1	range resample (per pulse, xM)	RES	SIG + COEF idx/wq	SCRATCH (row <code>invorder[i]</code> , tan ϕ -sorted)	resample
2	corner-turn (transpose)	CT	SCRATCH	SIG	resample
3	azimuth resample (per range bin, xNp)	RES	SIG + COEF	SCRATCH (uniform k-space)	resample
4	window (2-D Hamming)	WIN	SCRATCH + hamr/hamc	SCRATCH (in-place taper)	window
5	range FFT	FEED -> CoreFFT -> <code>fft_unloader</code>	SCRATCH (stream)	SCRATCH (AXI4 write)	rangeFFT
6	corner-turn (transpose)	CT	SCRATCH	SIG	cornerturn

#	Stage	Kernel	Reads	Writes	Timing stage
7	azimuth FFT	FEED -> CoreFFT -> fft_unloader	SIG (stream)	SIG (AXI4 write)	azFFT
8	detect (magnitude)	CPU (DET bypassed)	SIG	OUT (final image)	detect

Note that the corner-turn appears twice — once inside resample and once between the two FFT passes — and that steps 2 and 6 both write SIG while steps 1 and 3 both write SCRATCH. This is the SIG/SCRATCH ping-pong; a stage that reloads a buffer it just wrote is a bug, not an optimisation.

2.1 Resample (keystone / polar-format interpolation)

Two passes with a transpose between them, all inside `resample_2pass()`.

- Pass 1 (range): each real pulse row of SIG (N samples) is resampled to the padded width N_p and written to SCRATCH at its `tan_phi`-sorted row `invord[i]`, so SCRATCH ends up pulse-sorted. Padded rows $M..M_p-1$ are then zeroed.
- Transpose SCRATCH -> SIG so range bins become rows.
- Pass 2 (azimuth): each range-bin row (M sorted pulses) is resampled to M_p , leaving the resampled k-space in SCRATCH.

The interpolation contract is a two-tap linear gather:

```
out[i] = in[idx[i]] + (in[idx[i]+1] - in[idx[i]]) * wq[i] / 32768
```

equivalently `in[idx]·(1-w) + in[idx+1]·w` with $w = wq/2^{15}$. `idx` is an index into the source in its natural order; `idx = -1` means out-of-range and zero-fills.

Coefficients are computed just-in-time on the MSS, one line at a time, from the small per-pulse geometry arrays — precomputing the full grid would be ~768 MiB. The MSS double-buffers: while the fabric kernel gathers line `i` from bank `b`, the CPU fills bank `b+1` for line `i+1`. Coefficient generation is float on the CPU (`sar_resample_coeffs.c`, mirrors the host `interp_coeffs()` at corr 1.0); the interpolation itself is fixed-point in fabric.

Measured split per line: kernel-wait 78%, coefficient compute 20%, cache flush 2%. The stage is bound by fabric gather throughput, not by coefficient generation — the coefficient work is already hidden behind the kernel.

2.2 Window

Separable Hamming taper applied as the on-the-fly product `hamr[j] · hamc[k]` from two Q15 1-D tapers, rather than a materialized 2-D table. Zero inside the zero-pad region.

Do not attempt to fuse this into the resample gather. It was tried, was bit-exact in `shls sw` and `ll=1` in `shls hw`, and hit two distinct SmartHLS miscompiles on silicon — see `fpga/SMARTHLS_ANTIPATTERNS.md`.

2.3 Range FFT, corner-turn, azimuth FFT

See section 4 for the FFT engine and BFP contract.

The corner-turn is the load-bearing data-movement primitive: a global transpose cannot be fused into a neighbouring stage, so it is the one stage that must fully materialize the frame in DDR. It is tiled through on-chip LSRAM and is the reason the buffer plan needs a distinct destination (SCRATCH -> SIG rather than in place).

2.4 Detect

Per-pixel magnitude $\text{sqrt}(I^2 + Q^2)$, saturated to uint16.

This runs on the CPU, not in fabric. The fabric detect kernel is bypassed because SmartHLS mis-synthesized its sign extension: source-correct C `((int16_t)(x >> 16))` was read as unsigned in the generated RTL, so every negative-I pixel overflowed and saturated to 0xFFFF — about half the image — collapsing correlation. Both `shls` cosim and a correlation check passed anyway; only a value-level comparison caught it.

I and Q must be read as signed int16 before squaring. The branchless reference formula is:

```
sext16(u) = (int32_t)((u & 0xFFFF) ^ 0x8000) - 0x8000
```

Engine selection is runtime: `detect_mode @ 0xB0059118`, 1 = CPU detect (the shipping path).

3. Fixed-point and data contracts

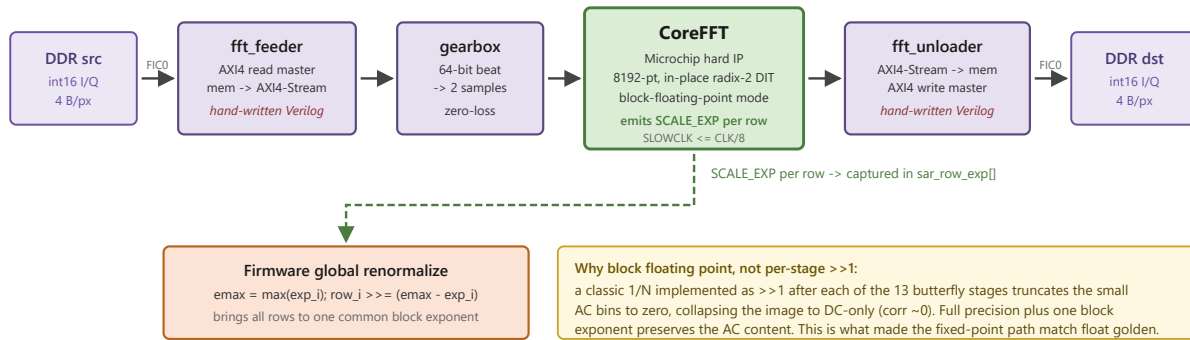
- Complex samples are int16 I and int16 Q packed as one 32-bit word per pixel, hi16 = I, lo16 = Q.
- The detected OUT image is uint16 magnitude, 2 bytes per pixel.
- Therefore a complex buffer is 4 B/px and OUT is 2 B/px, and **the same byte offset addresses different rows in the two**. Always compute row addresses explicitly as `base + row * GRID * bytes_per_px`. This has bitten the project more than once.
- Resample weights `wq` are Q15 (`w = wq / 32768`).
- Window tapers `hamr`, `hamc` are Q15 int16.
- Geometry arrays (`f0`, `df`, `pr`, `tans`, `krgrid`, `kcgrid`) are float32; `invorder` is int32.

Datapath sizing, from the fixed-point emulator study: 16-bit mantissa, 18-bit twiddle, 48-bit accumulate, BFP arithmetic shift after every stage. Measured FFT block-exponent growth is +9/+10 bits, covered by a 5-bit exponent per FFT line. At 16-bit the fixed image is visually identical to float (corr 0.9992 on the ship scene) and retains ~53 dB usable dynamic range at ~4.6 ENOB. If 53 dB proves marginal, 18-bit mantissa buys roughly 12 dB more and still fits one 18x18 DSP per multiply.

4. FFT engine and block floating point

Fabric FFT chain — CoreFFT hard IP (the shipping FFT path)

Selected at runtime by SAR_FFTMODE @ 0xB005_9110 = 1. Mode 0 is the legacy CPU FFT in sar_fft.c.



Note: the IP's SCALE_EXP register is NOT the same quantity as a software BFP block exponent (the IP applies an ~unconditional 1/N scale). Do not compare the two directly.

The feeder and unloader are hand-written Verilog because SmartHLS synthesizes mem <-> stream kernels to dead RTL on this toolchain (mem -> mem HLS kernels are fine).

An earlier HLS FFT (hls_fft/fft_in_place_bfp) dropped the twiddle term in generated RTL on silicon while every C-simulation passed at corr 0.9999 — it is not used. See SAR_PIPELINE_STATUS.md.

Re-arm hazard: the golden testbench only ran one transform. The MEMBUF overwrite and READ_OUTP/DATA0-latency traps appear only on the second and later frames.

The shipping FFT is the Microchip CoreFFT hard IP, driven as a streaming chain `fft_feeder -> gearbox -> CoreFFT -> fft_unloader`. It is selected at runtime by `SAR_FFTMODE @ 0xB0059110 = 1`, which is what the pipeline runs with; mode 0 is a retained legacy CPU FFT in `sar_fft.c`.

The feeder and unloader are hand-written Verilog. This is not a style choice: SmartHLS synthesizes mem-to-stream and stream-to-mem kernels to dead RTL on this toolchain. Memory-to-memory HLS kernels are fine.

Block floating point is the reason the fixed-point pipeline matches the float golden. CoreFFT runs in BFP mode and reports a per-row `SCALE_EXP`; the true value is `DATA0 * 2^SCALE_EXP`. Firmware captures the per-row exponents in `sar_row_exp[]` and renormalizes globally to a common block exponent, `row_i >>= (emax - exp_i)`.

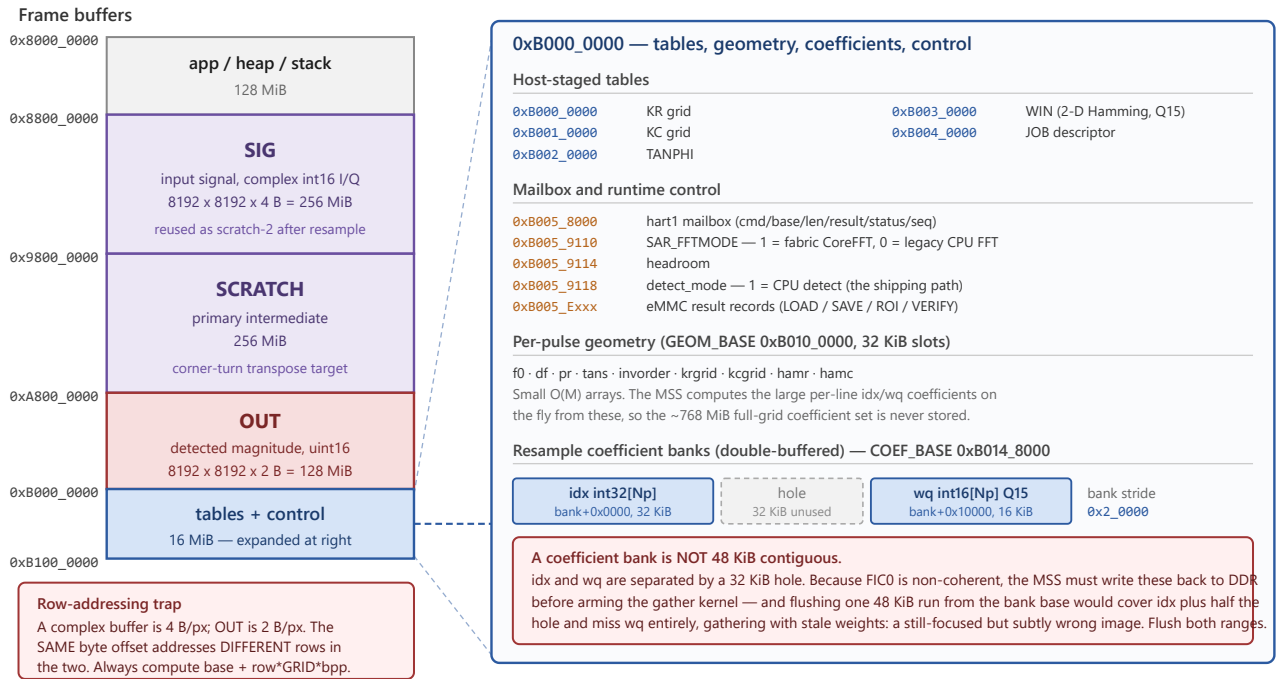
The alternative — a classic 1/N implemented as `>>1` after each of the 13 butterfly stages — truncates the small AC bins to zero and collapses the image to DC-only (corr ~0). Fixed-point FFT dynamic range must be managed with a block exponent, not per-stage truncation.

Note that the IP's `SCALE_EXP` register is not the same quantity as a software BFP block exponent (the IP applies an approximately unconditional 1/N scale). Do not compare the two directly.

5. Memory map

DDR memory map — 1 GiB LPDDR4, cached window at 0x8000_0000

Single source of truth: `src/sar/ddr_sar_layout.h`, mirrored by `mpfs/host/ddr_layout.py`. Keep the two in lock-step.



All addresses are physical, in the cached DDR window. Fabric masters reach DDR over FIC0 and do not snoop L2 — see SAR_DESIGN.md "Cache coherency" for the required flush at each handoff.

Defined in `src/sar/ddr_sar_layout.h` and mirrored by `mpfs/host/ddr_layout.py`. These two must stay in lock-step; the header is the bare-metal mirror, the Python module is the host-side single source.

Base	Size	Region
0x8000_0000	128 MiB	app / heap / stack
0x8800_0000	256 MiB	SIG — input signal, complex int16 I/Q; reused as scratch-2 after resample
0x9800_0000	256 MiB	SCRATCH — primary intermediate, corner-turn target
0xA800_0000	128 MiB	OUT — detected magnitude, uint16
0xB000_0000	16 MiB	tables, geometry, coefficient banks, mailbox and control

Buffers ping-pong SIG <-> SCRATCH so an in-place FFT never reads and writes the same page.

Within the tables region: host-staged grids at 0xB000_0000 (KR, KC, TANPHI, WIN, JOB); the hart1 mailbox at 0xB005_8000; runtime knobs at 0xB005_9110/9114/9118; eMMC result records at 0xB005_Exxx; per-pulse geometry from GEOM_BASE 0xB010_0000 in 32 KiB slots; and the double-buffered resample coefficient banks from COEF_BASE 0xB014_8000, stride 0x2_0000.

A coefficient bank is not 48 KiB contiguous: **idx** (int32[Np], 32 KiB) sits at bank+0x0000 and **wq** (int16[Np], 16 KiB) at bank+0x10000, with a 32 KiB hole between. Anything that publishes a bank must treat it as two disjoint ranges.

6. Cache coherency

FIC0 is not coherent — fabric masters reach DDR without snooping the L2. Every handoff across the CPU/fabric boundary therefore needs an explicit cache operation:

Handoff	Requirement
CPU writes data, fabric reads it	Write back L2 to DDR before arming the kernel
Fabric writes data, CPU reads it	Invalidate L2 before reading, or the CPU sees stale lines
SDMMC DMA writes DDR, CPU or fabric reads it	Write back / invalidate after the transfer completes

Two mechanisms are in use:

- `flush_l2_cache(hartid)` — the MPFS HAL's whole-cache operation. It is a way-by-way walk: for each of the 16 ways it reads 131 KiB from the L2 zero device (~268k volatile loads) and manipulates the WayMask allocation policy. Correct, but expensive, and it invalidates everything.
- `flush_coef_bank_to_ddr(bank, n)` in `sar_sequencer.c` — writes only the covering lines to the CCACHE FLUSH64 register (`CACHE_CTRL_BASE 0x0201_0000`), which writes back and invalidates the line containing a given physical address. Used on the resample per-line critical path, where the only CPU-dirty data is the coefficient bank. Roughly 768 stores instead of ~268k loads.

Because the coefficient bank is discontinuous (section 5), `flush_coef_bank_to_ddr()` flushes `idx` and `wq` as separate ranges. A single 48 KiB run from the bank base would cover `idx` plus half the hole and miss `wq` entirely, so the kernel would gather with stale weights — producing a still-focused but subtly wrong image that a correlation check would likely not catch.

Known gap: `MBX_CMD_CRC32`'s handler writes `mbx->result` with only a fence and no writeback, so a JTAG physical read of it returns a stale value. The dedicated result records at `0xB005_Exxx` are written back and are trustworthy.

7. On-board data path (eMMC)

The scene lives on the board's soldered 8 GB eMMC, so a run needs no host data transfer. This retires the ~3 h JTAG scene load that previously preceded every run.

Two fixed-LBA partitions, each self-describing via a superblock and TOC, so job geometry is reconstructed at load time and never trusted from volatile RAM:

Region	LBA	Size	Contents
INPUT (<code>SARI</code> , 0x53415249)	<code>0x80000</code> (256 MiB)	4 GiB	packed scene image: superblock + one blob per scene
OUTPUT (<code>SARO</code> , 0x5341524F)	<code>0x880000</code> (4.25 GiB)	3 GiB	superblock + persisted output image(s)

A scene blob (`SARB`, 0x53415242) is a header, a segment table, and ten 512-aligned role segments: `sig`, `f0`, `df`, `pr`, `tans`, `invorder`, `krgrid`, `kcgrid`, `hamr`, `hamc`. Each segment is tagged by role and the firmware resolves role to DDR address via `sar_emmc_role_addr()`, scattering to the fixed addresses in section 5. The JOB descriptor carries only scalars (M, N, `fft_r`, `fft_a`, `out_dtype`, `bfp_in_exp`, `sig_len`, `sig_crc`) plus the SIG/OUT/SCRATCH bases — the pipeline reads geometry from the fixed addresses, not from the JOB's legacy address fields.

Measured rates (LEGACY mode, 25 MHz, 8-bit, single-block): read 1.5 MB/s, write 0.13 MB/s. Scene LOAD is 78 s; persisting the 128 MiB OUT image is about 16 min.

What eMMC does and does not solve: it eliminates the recurring 3 h JTAG input load, which is the win. It does not speed host transfer — dumping the full OUT image to the PC is still bound by the FlashPro6 JTAG link at roughly 9 KB/s. Keep the image on-card and dump small ROI crops to verify; a faster offload needs a different host link, not a faster card.

SDMA would speed writes by one to two orders of magnitude but is currently unusable: hart1 runs with `MSTATUS_MIE` cleared and `MSS_MMC_sdma_write` completes via the MMC PLIC ISR, so with interrupts off it spins forever in an unhaltable hang. Single-block transfers are synchronous and cannot hang.

Output persistence is commit-last: invalidate the superblock, write the image, write the superblock last. A power loss mid-image therefore leaves an invalid superblock and readers reject the torn image. Do not reorder this.

8. Control interface

The hart1 mailbox at `0xB005_8000` is the single entry point. Layout: `cmd` +0, `base` +4, `len` +8, `result` +0xC, `status` +0x10 (done = `0xC0FFEE03`), `seq` +0x14. Write `base` and `len` first and `cmd` last; the hart clears `cmd` to acknowledge.

Command	Code	Purpose
ELOD	0x454C4F44	Load scene eMMC -> DDR, plus JOB
PIPE	0x50495045	Run the image-formation pipeline
ESAV	0x45534156	Persist OUT -> eMMC SARO
EVOU	0x45564F55	Verify the persisted image by full CRC against the TOC
EROI	0x45524F49	Crop an ROI from the DDR OUT image
EROE	0x45524F45	Crop an ROI from the persisted SARO image
EPRV	0x45505256	Provision the INPUT partition
EMMC	0x454D4D43	eMMC self-test

Results are latched in dedicated DDR records (`0xB005_E000` LOAD, `E100` SAVE, `E200` ROI, `E300` VERIFY, `0xB005_D000` provision), each written back to DDR so a JTAG physical read sees them. Verdict 0 is pass; 1 PARAM, 2 INIT, 3 MAGIC, 4 IO, 5 CRC.

Per-stage timing is instrumented in `sar_stage_ts[0..6]` from MTIME at 1 μ s per tick, readable without re-running the pipeline via `bash mpfs/host/run_stage_timing.sh`.

Fabric kernels are controlled over AXI4-Lite through FIC0; per-kernel register offsets are in `src/sar/sar_kernels.h`. Note there are two register-map models in the project's history — the hardware uses the per-kernel model in `sar_kernels.h`, not the older monolithic one.

9. Verification contract

Correlation is scale-, phase- and orientation-invariant, so it hides real bugs — the detect sign-extension defect passed a correlation check while corrupting half the image. Verify by value: feed known inputs, diff actual complex sample values against the bit-accurate emulator ([silicon_emulator.py](#), which equals the float golden at corr 1.0), and only then compare to the reference image.

When comparing a board image to the golden, account for orientation before declaring a divergence. The board result matches the golden in the `T.rot180` orientation (`board == golden.T[::-1,::-1]`). A naive band comparison once read corr 0.06 on a correct image that scanned to 0.97 — run the full 8-dihedral orientation search ([mpfs/host/correlate_cpufft.py](#)) first.

Current reference result: corr 0.9923 against [golden_small_mag.npy](#) on the Centerfield decimated scene, with a point-target crop at 0.9962.

10. Known deviations from the ideal design

These are deliberate and load-bearing. Do not "fix" them without reading the linked history.

Deviation	Reason
Detect runs on the CPU, not fabric	SmartHLS sign-extension miscompile; costs 19.7 s
FFT feeder/unloader are hand-written Verilog	SmartHLS mem <-> stream kernels synthesize to dead RTL
Window is a separate pass, not fused into resample	Two distinct SmartHLS miscompiles on fusion
eMMC uses single-block transfers, not SDMA	Interrupts are off on hart1; SDMA would hang unhalatably
Fabric runs at 62.5 MHz	75 MHz effort stopped at marginal ROI; the HLS window/resample kernels are the Fmax limiter, not CoreFFT or the DDR bridge
~50% of OUT saturates at 65535	Detect BFP shift headroom; cosmetic, correlation is measured on unsaturated pixels

SmartHLS is treated as an untrusted, behavioural-only tool. Every kernel output is value-checked on silicon after a rebuild — see the [hls-trust-harness](#) skill and [fpga/SMARTHLS_ANTIPATTERNS.md](#).